



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
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Lab Report NO: 02
Course Title: Data Communication Lab
Course Code: CSE 312 **Section:**232 D2

Lab Experiment Name: Error Detection & Correction Using Hamming Code

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Lab Report Status

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Comments:..... **Date:**.....

TITLE OF THE LAB EXPERIMENT

Error detection and correction using the Hamming code.

Problem Statement

In digital communication, data may become corrupted because of noise or interference during transmission. In this task, a 7-bit Hamming (7,4) encoded message is received, where at most one bit may contain an error. The program is required to detect the error, correct it if one exists, display the corrected 7-bit code, and recover the original 4-bit data from the corrected message.

IMPLEMENTATION

Here is the coding part of this task :

```
#include <stdio.h>

int main() {
    char code[8];
    int h[8];
    int s1, s2, s4, error;
    scanf("%s", code);
    for (int i = 1; i <= 7; i++) {
        h[i] = code[i - 1] - '0';
    }
    s1 = h[1] ^ h[3] ^ h[5] ^ h[7];
    s2 = h[2] ^ h[3] ^ h[6] ^ h[7];
    s4 = h[4] ^ h[5] ^ h[6] ^ h[7];
    error = s1 + (s2 * 2) + (s4 * 4);
```

```

if (error == 0) {
    printf("No error detected.\n");
} else {
error); printf("Error detected at position %d.\n",
    h[error] = !h[error];
}
printf("Corrected codeword: ");
for (int i = 1; i <= 7; i++) {
    printf("%d", h[i]);
}
printf("\nOriginal data: ");
printf("%d%d%d%d\n", h[3], h[5], h[6], h[7]);
return 0;
}

```

OUTPUT



```

cd "c:\Users\Hp\Downloads\"
.\Hamming_report }

Please Enter you 7 bit binary string: 1111010

Error detected at position 2.
Corrected codeword: 1011010
Original data: 1010
PS C:\Users\Hp\Downloads>

```

Fig. 1: Output for bitstream 1111010

```
PS D:\Ehealthbd> cd "c:\Users\Hp\Downloads\" ; if  
_report }  
  
Please Enter you 7 bit binary string: 1011010  
  
No error detected.  
Corrected codeword: 1011010  
Original data: 1010  
PS C:\Users\Hp\Downloads>
```

Fig. 2: Output for bitstream 1011010

ANALYSIS AND DISCUSSION

This experiment demonstrated how the Hamming (7,4) code can detect and correct a single-bit error during data transmission. The program successfully identified whether an error existed, corrected the incorrect bit, and recovered the original 4-bit data from the received 7-bit code. Through this experiment, I learned how parity bits are used for error detection, how the error position is calculated using the syndrome, and how Hamming code improves the reliability of digital communication by ensuring accurate data transmission.